

1 What is a reasonable estimate of the weight of an adult human?

- A** $7 \times 10^0 \text{ N}$ **B** $7 \times 10^2 \text{ N}$ **C** $7 \times 10^4 \text{ N}$ **D** $7 \times 10^6 \text{ N}$

2 What describes a set of data with a high precision?

- A** data measured using equipment with small scale divisions
B data that is close to the accepted value
C data with each value having a low uncertainty
D data with repeats that are close to each other

3 Which physical quantity could have units of $\text{N s}^2 \text{ m}^{-1}$?

- A** acceleration
B force
C mass
D momentum

4 A ball is released from rest. The distance the ball falls and the time the ball takes to fall that distance are both measured.

The percentage uncertainty in the measured distance is negligible. The percentage uncertainty in the measured time is 4%.

The distance and the time are then used to calculate the acceleration of free fall.

Air resistance is negligible.

What is the percentage uncertainty in the calculated value of the acceleration of free fall?

- A** 2% **B** 4% **C** 8% **D** 16%

5 A car has an initial velocity u . The car then moves with constant acceleration a in a straight line through a displacement d . The car reaches a final velocity v .

Which expression gives the initial velocity u of the car?

- A** $v - ad$ **B** $\sqrt{v^2 - 2ad}$ **C** $v + ad$ **D** $v - \frac{1}{2}ad^2$

6 A bicycle brakes so that it undergoes uniform deceleration from a speed of 8 m s^{-1} to 6 m s^{-1} over a distance of 7 m.

The deceleration of the bicycle remains constant.

Which further distance will the bicycle travel before coming to rest?

- A** 7 m **B** 9 m **C** 16 m **D** 21 m